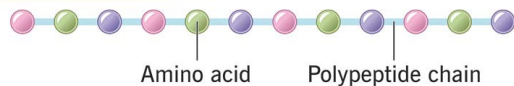


A brief introduction on Protein Structure Prediction and Protein Language Models

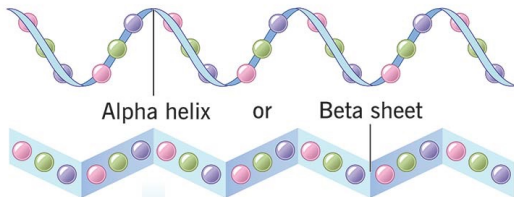
Amino acid sequences is **folded** to form a complex structure

Proteins

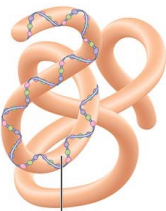
Primary structure



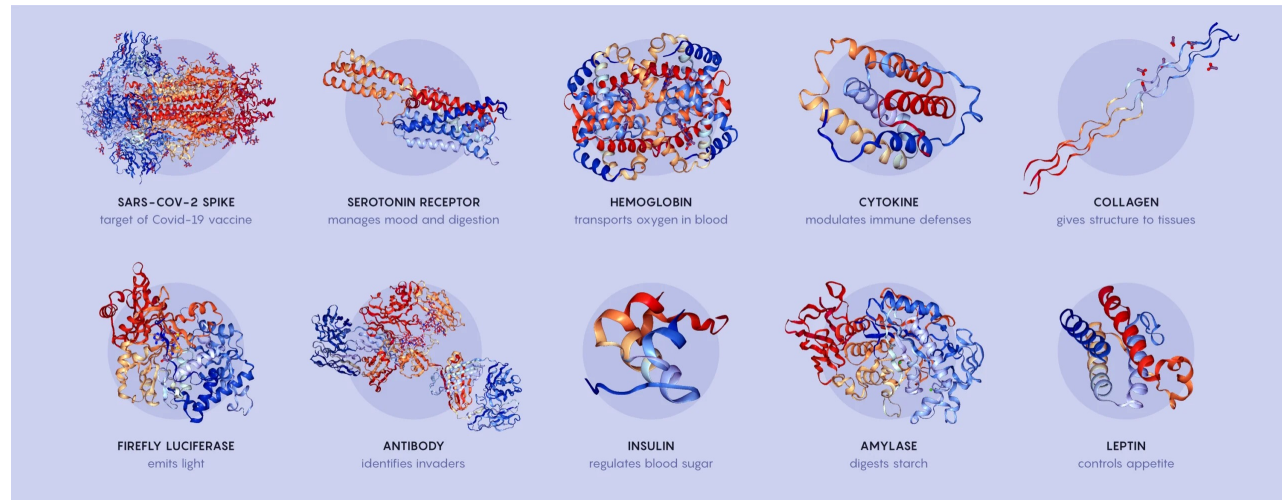
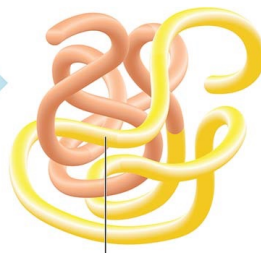
Secondary structure



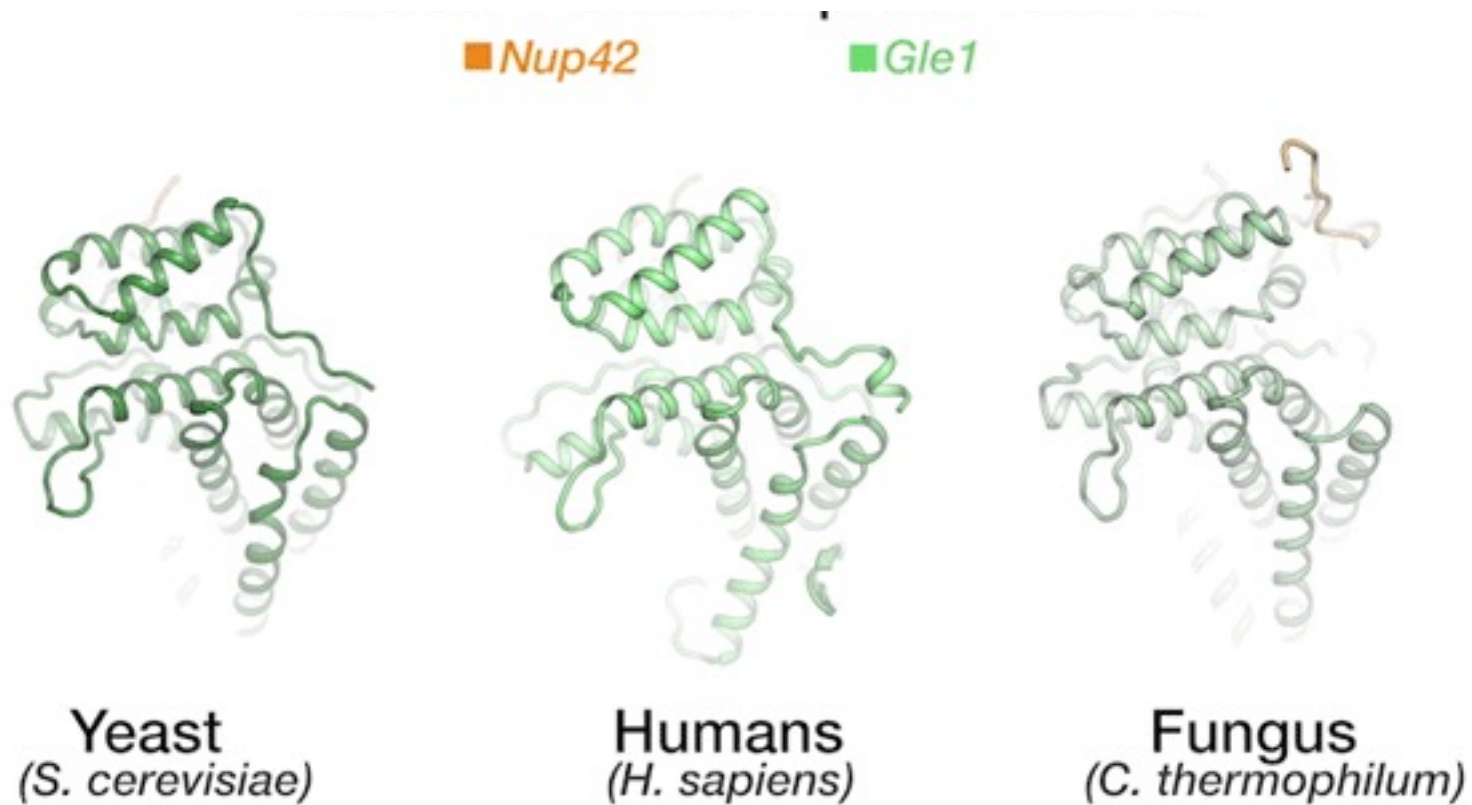
Tertiary structure



Quaternary structure

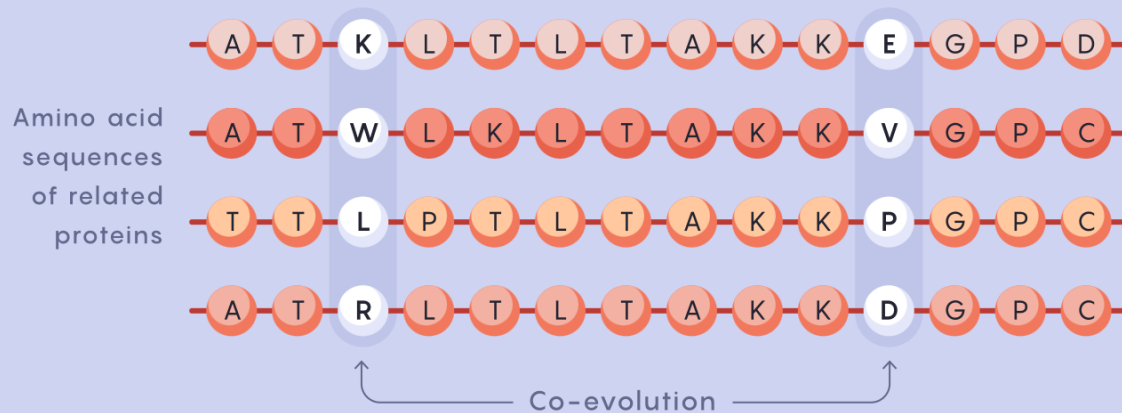


Nature reuses solutions and structures

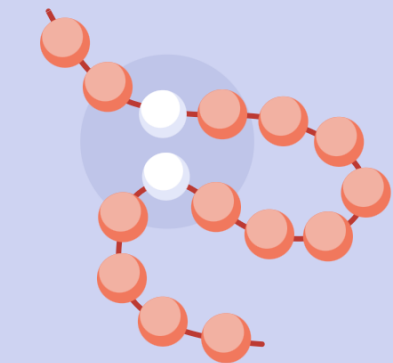


Co-Evolution leads to conserved structure

Protein Co-Evolution Reveals Structure

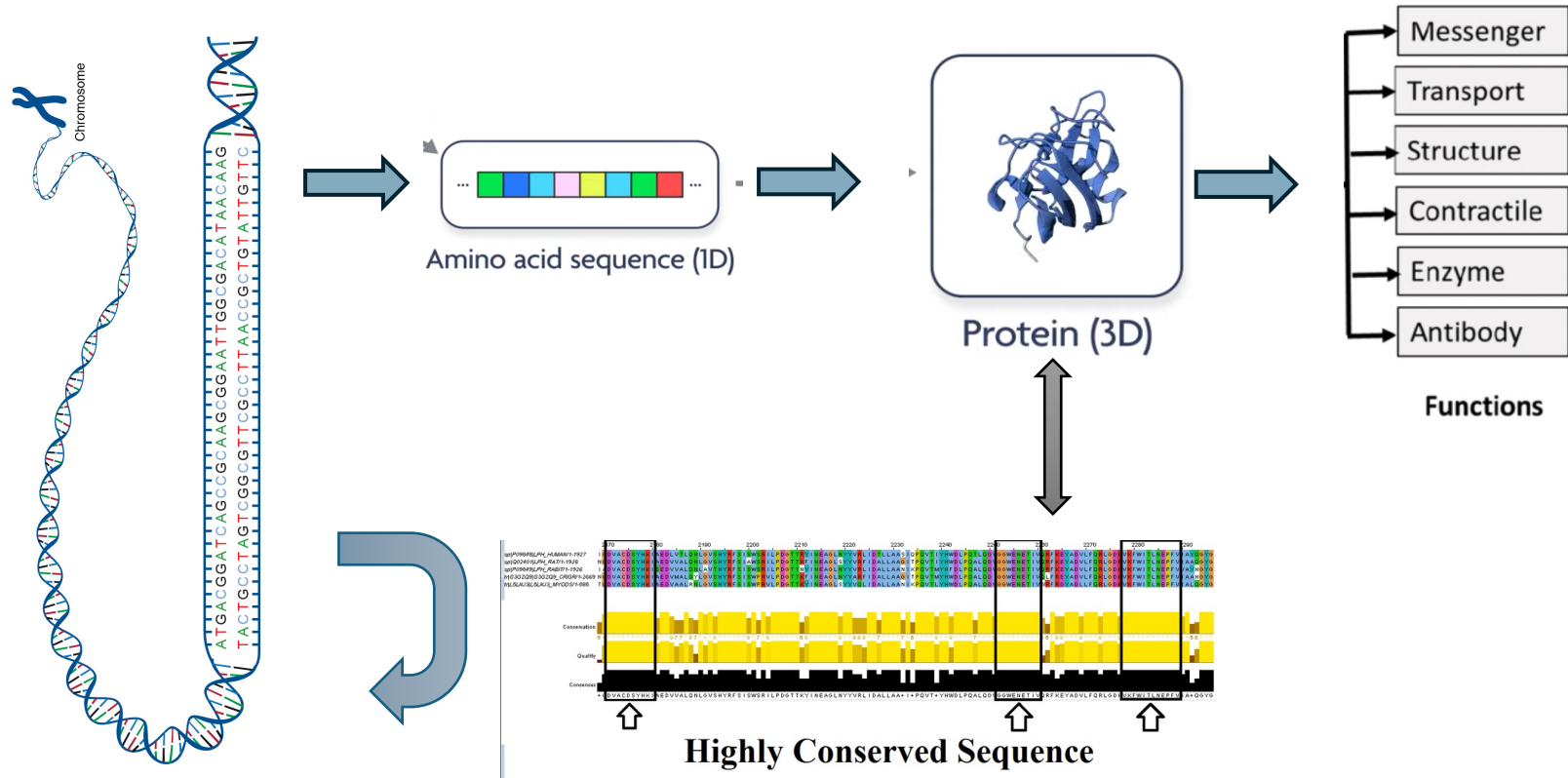


Scientists can compare the amino acid sequences of hundreds or thousands of related proteins to identify amino acid positions that mutated together — or “co-evolved” — across multiple polypeptides.



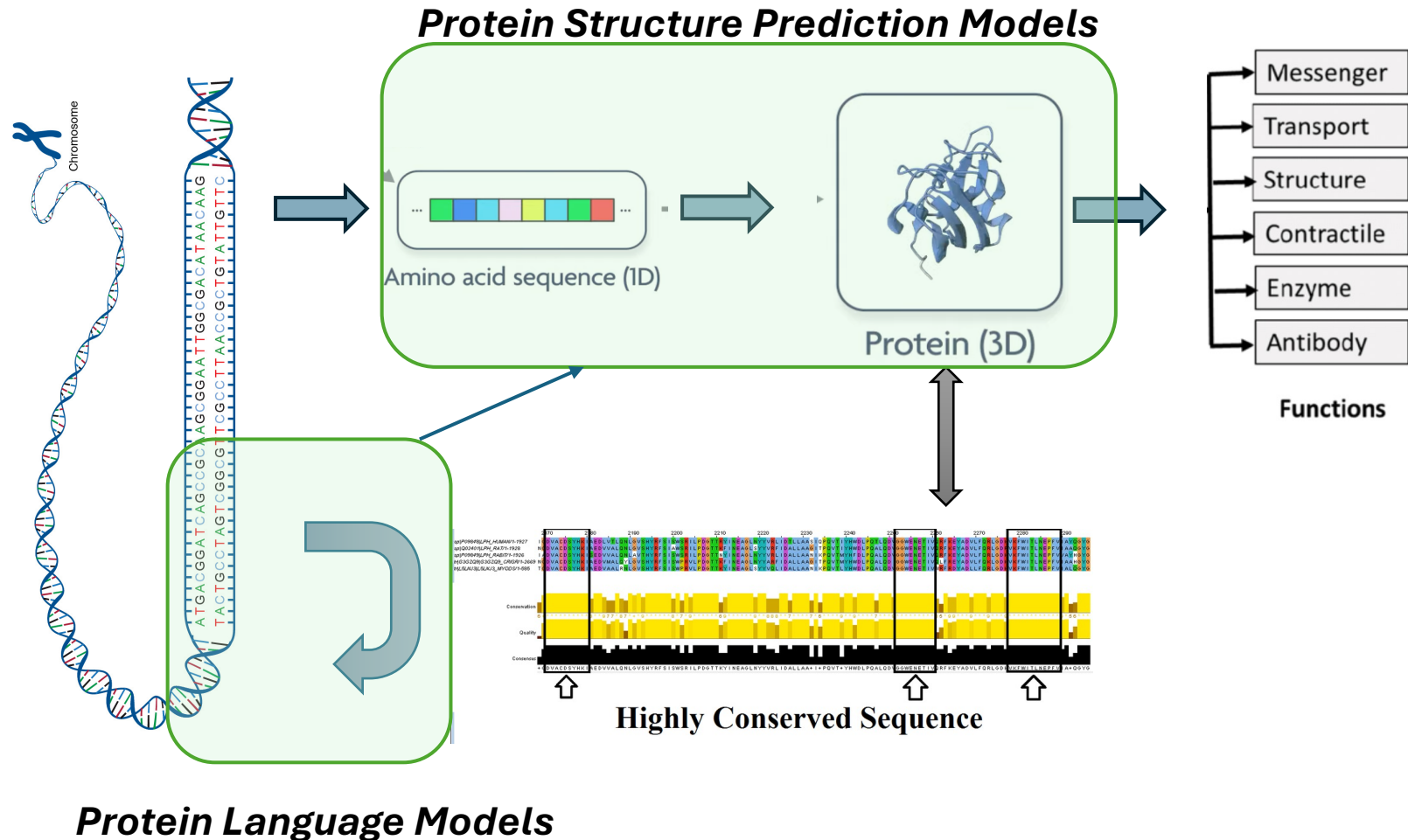
Those amino acids are likely to be close together in the polypeptide’s folded form.

What is the big challenge ?

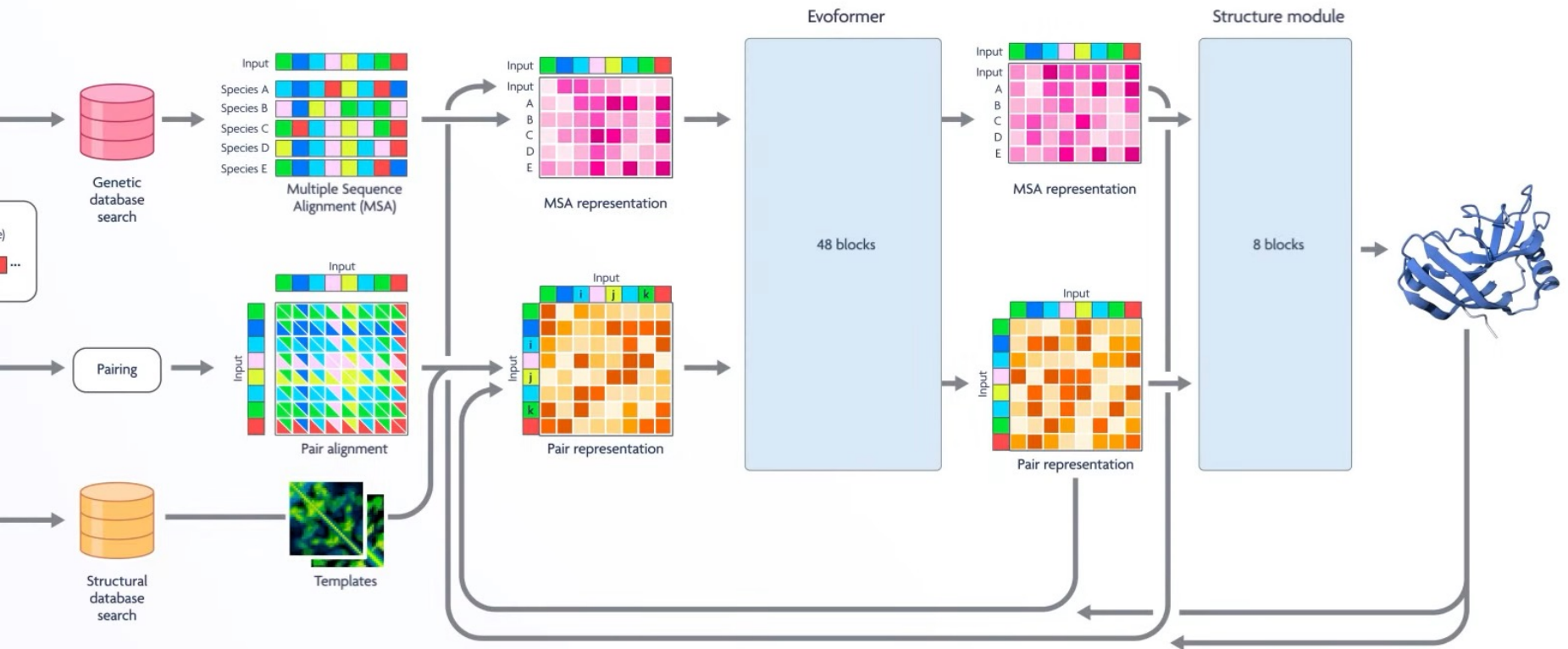


AI models in protein models :

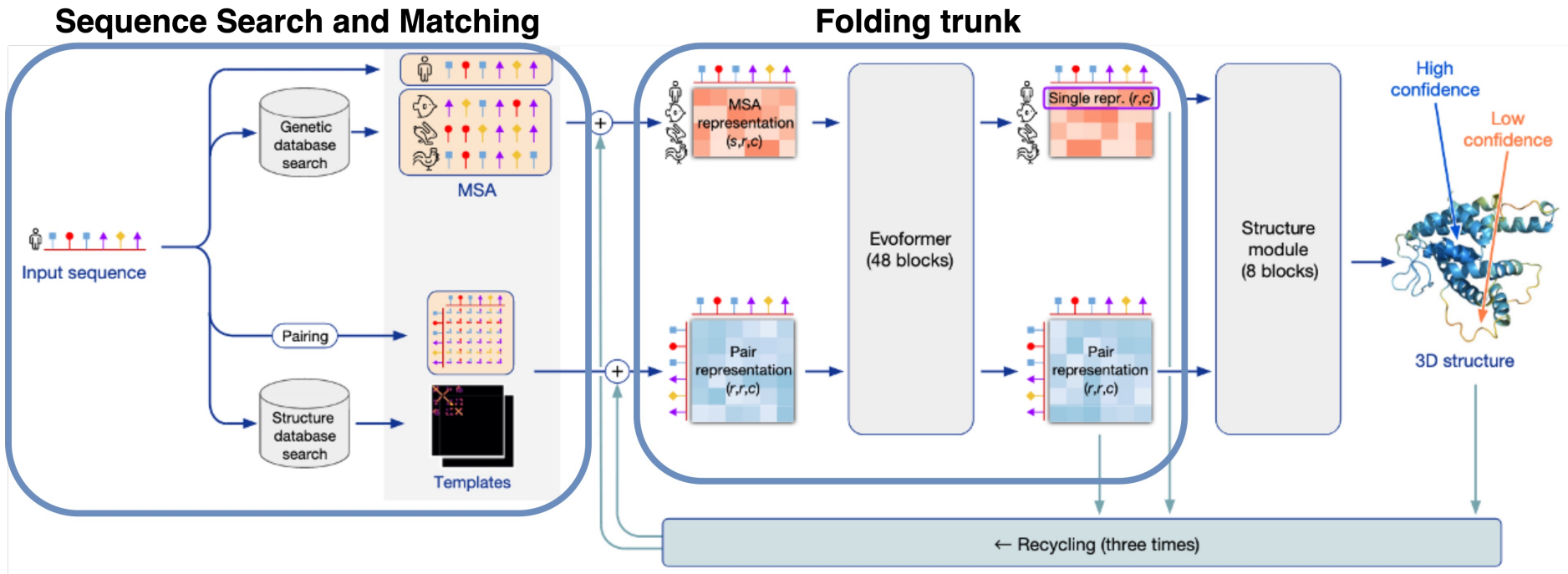
Two main category



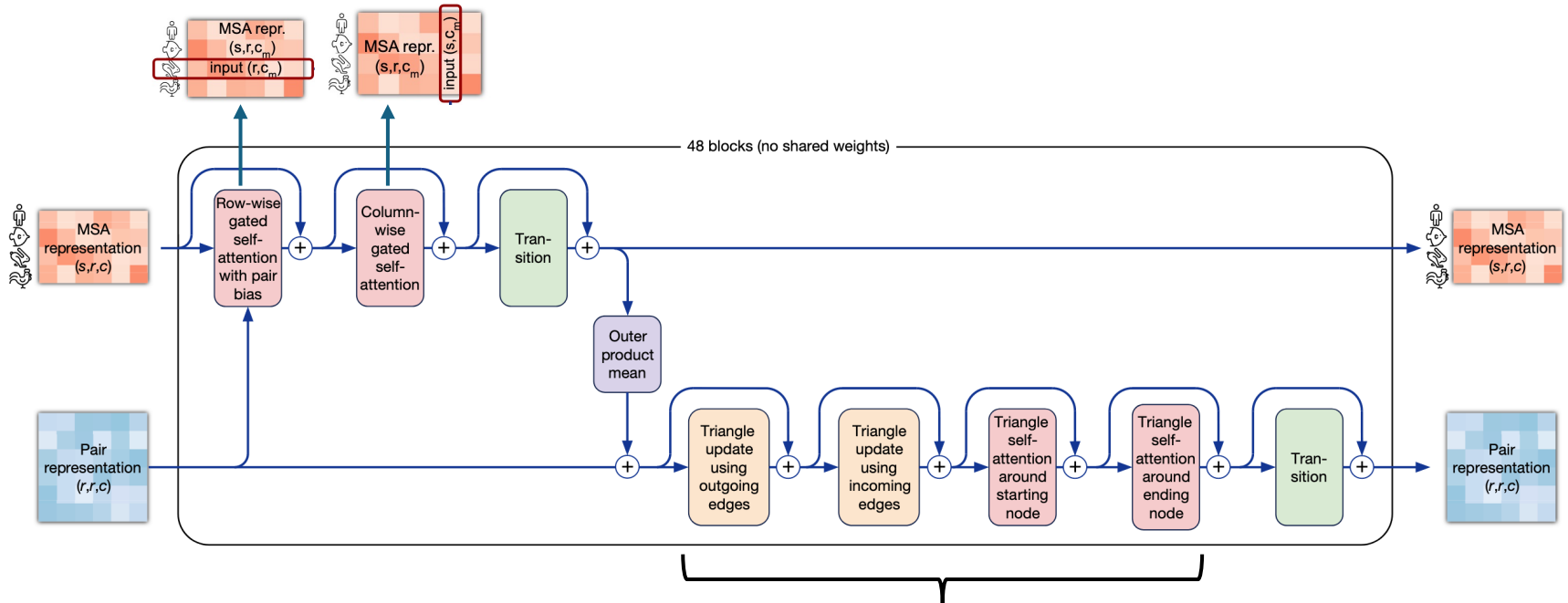
AlphaFold 2 Architecture



A closer look at AlphaFold2 architecture

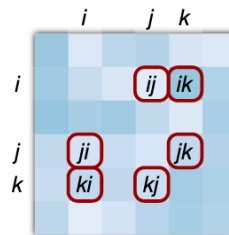


A closer look at folding trunk

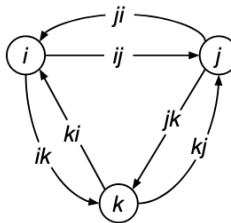


Triangle attention applies the triangle inequality constraint

Pair representation
 (r, r, c)



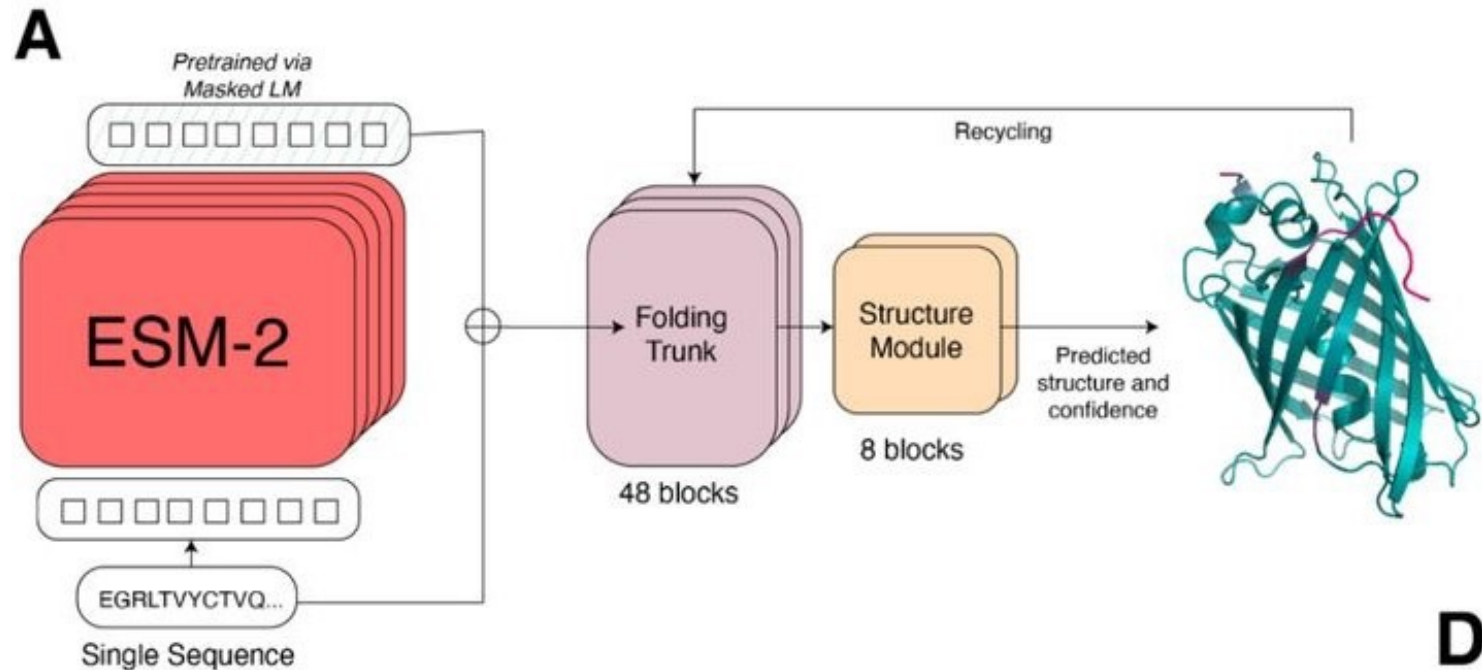
Corresponding edges
in a graph



Triangle inequality
theorem

$$ij + ik \geq jk$$

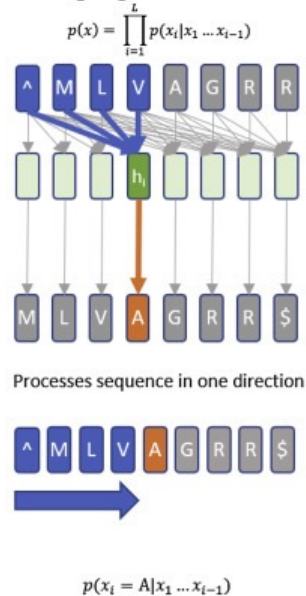
What about Protein Language Models ?



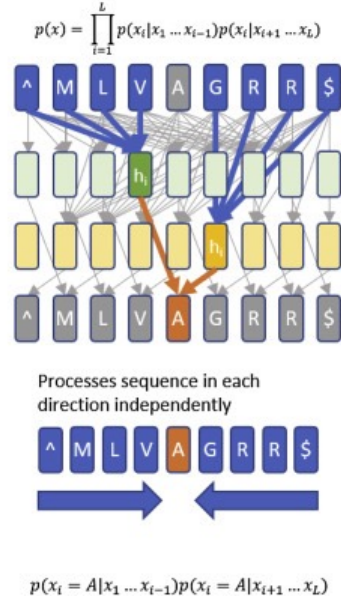
How are PLM trained ?

PLM embeddings have information about protein functions

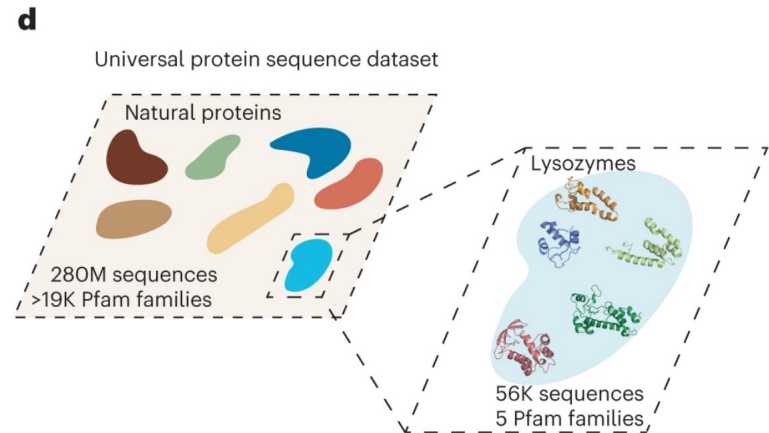
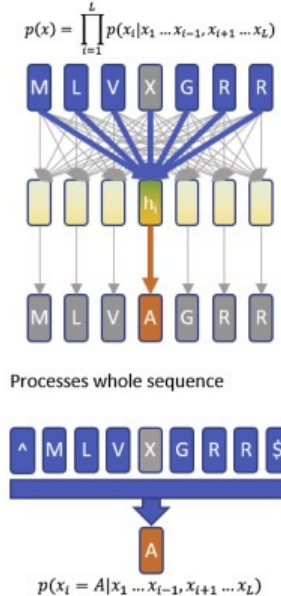
A Autoregressive language model



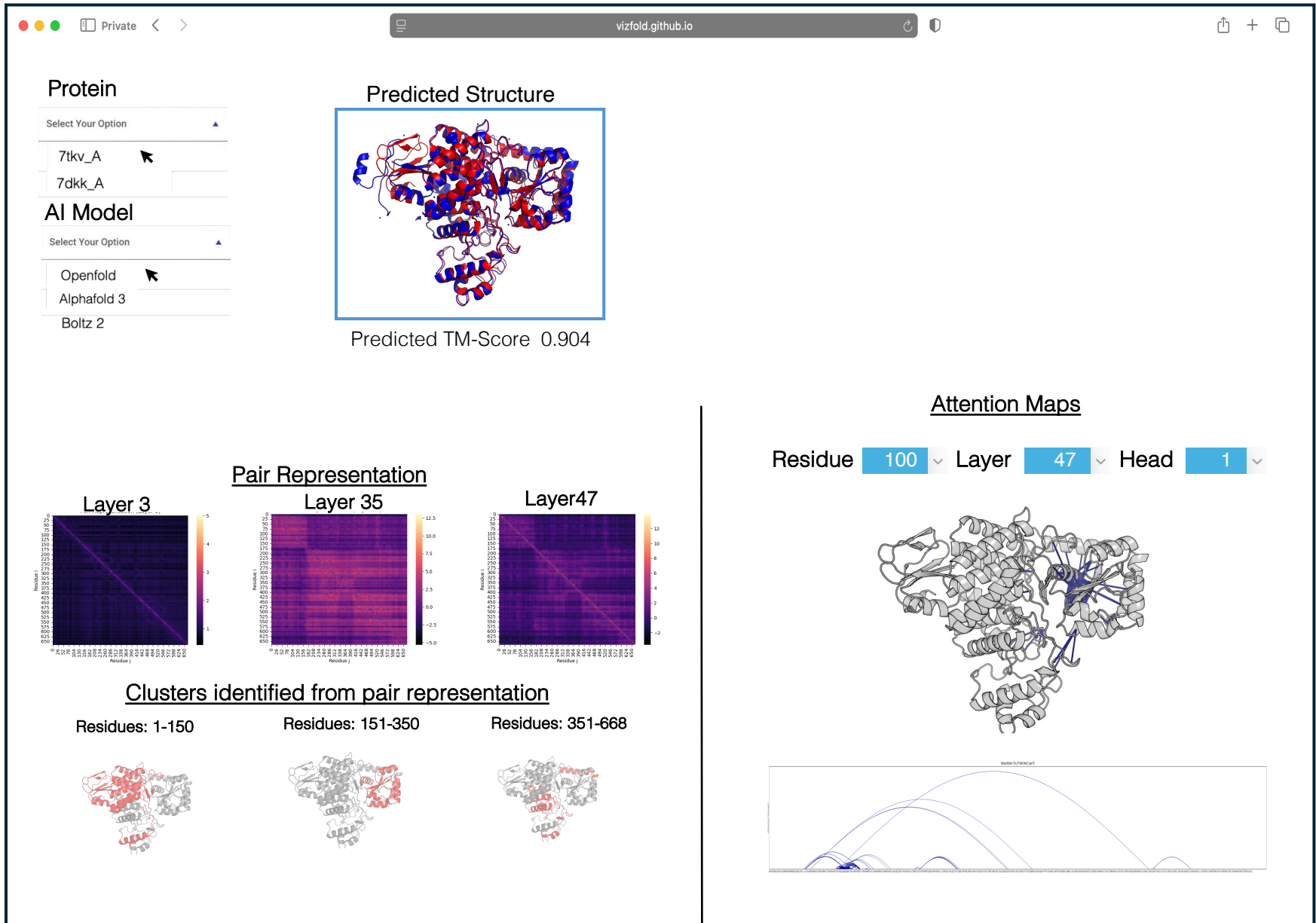
B Bidirectional language model



C Masked language model



Goal : Web Portal for **Inference and Interpretability** **Trace** for AI-based Protein Models



Slides by Devin Fromond

TA for Apache Airavata VIP Class

Refer: <https://www.tamarind.bio/>

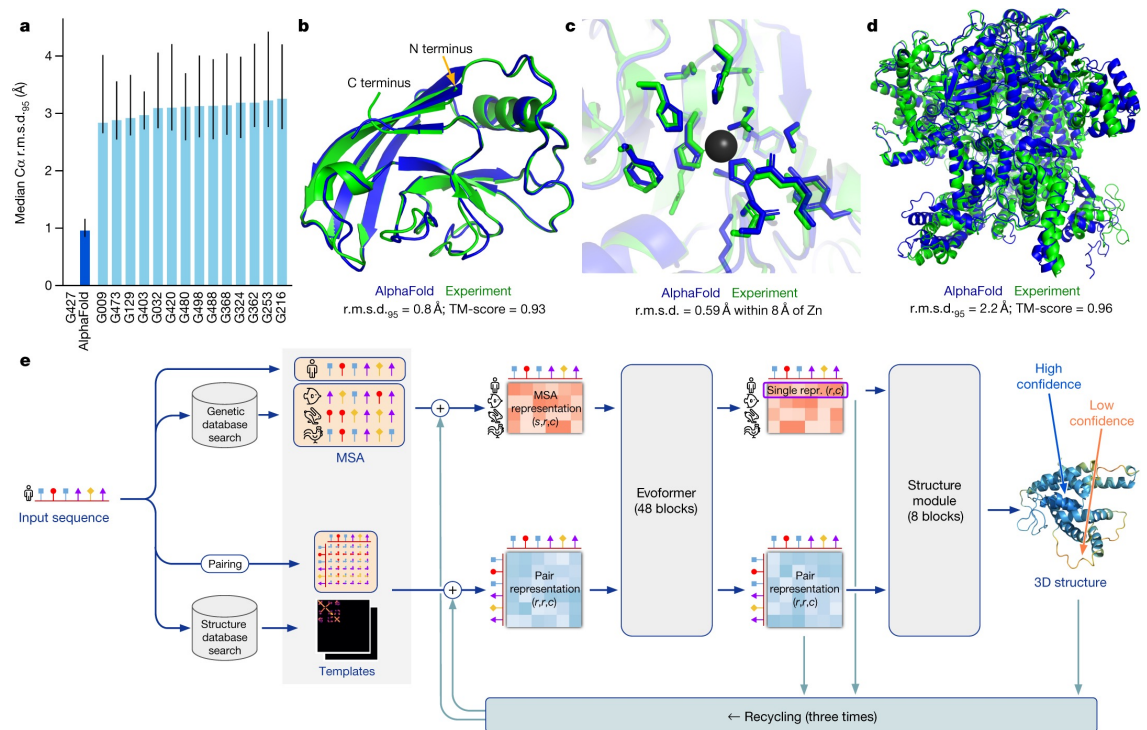
Toward Modular and Interpretable Protein AI Pipelines

Apache Airavata | VIP VYL

Vizfold Overview

Instead of treating the model as a black box, VizFold tries to expose the intermediate representations inside the model, such as:

- embeddings
- attention maps
- intermediate geometric structures



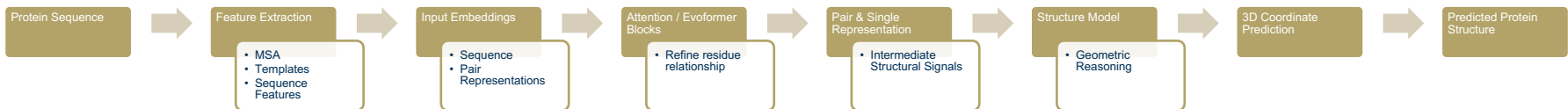
What Could be more Interesting...

Current VizFold Pipeline



VizFold lets us inspect some intermediate signals, but the pipeline is mostly linear and fixed.

Future VizFold Pipeline



Researchers could: inspect intermediate outputs, modify parameters, rerun downstream stages, compare results.

Two Directions that Enable This

Modularization Intermediate Layers

Expose pieces of the pipeline so they can be run independently:

```
run_embeddings()  
run_attention()  
run_structure()
```

This would allow intermediate visualization, partial pipeline execution, experimentation with parameters.

Standardizing Module Inputs/Outputs

Define a consistent interface between modules:

```
module(input_data, parameters) → output data
```

This makes it easier to plug in new tools, experiment with intermediate states, and test modifications.

Why this is Interesting

This type of workflow is common in real research and industry platforms.

Instead of running models, scientists want to:

1. Run part of the model
2. Inspect intermediate signals
3. Modify parameters
4. Continue Pipeline

Making VizFold modular would support this type of exploration.

AlphaFold   

- Predict the structure of a protein monomer or multimer given its amino acid sequence
- Uses DeepMind's AlphaFold v2.3.2 (implemented by ColabFold)
- Specify optional additional settings including recycles, MSA depth, custom templates
- Receive up to five pdb models, including scores for pTM, PAE, iPTM/iPAE (multimers), and pLDDT

Est. run time: 45m · [Show Outputs](#) [Show Settings](#) [Show Example Results](#)

Job Name

alphafold-example-6q2t

Sequence

Protein amino acid sequence, use : to separate chains for multimers

Sequence

Multiple

ID

Fasta

PDB

CSV

Excel

1 Sequence

 My proteins

MALKSLVLLSLLVLLLVRVQPSLGKETAAKFERQHMDSSSTAASSSNYCQMMKSRNLTK
DRCKPVNTFVHESLADVQAVCSQKNVACKNGQTNCYQSYSTMSITDCRETGSSKYPNCAYK
TTQANKHIIIVACEGNPYVPVHFDASV

Estimated Runtime 

Elapsed time: **7 min**

Load example inputs

View Optional Settings

▼

<https://app.tamarind.bio/tools/alphafold>